

DAY 1

SKILLBUILDER 3

Attention Mechanism in Large Language Models (LLMs)

Task 1: Concept Understanding

1. What is Attention?

Attention is a mechanism that allows a Large Language Model (LLM) to focus on the most relevant words or parts of a sentence when understanding or generating text. Instead of treating every word equally, the model assigns greater importance to words that provide the most useful context.

2. Why is Attention Important?

Attention helps LLMs understand context, resolve references between words, capture long-range relationships, improve accuracy, and generate more natural, meaningful responses.

Task 2: Applied Example

Sentence: "The bank will not approve the loan because it is risky."

1. Which word does "it" refer to?

The word "it" refers to "the loan". The sentence means that the loan is considered risky, so the bank does not approve it.

2. Explain how attention helps in this understanding

The attention mechanism compares the word 'it' with other words in the sentence and determines that 'loan' is the most relevant reference based on grammar, meaning, and context. This enables the model to correctly interpret the sentence.

Task 3: Use Case Thinking

1. Explain how attention improves:

Translation Systems

Attention helps translation models match words and phrases between languages, even when sentence structures differ. This produces more accurate and natural translations.

Chatbots

Attention enables chatbots to remember important details from the conversation, understand user intent, and provide context-aware responses instead of isolated answers.